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Repurposing 3D Generative Model for Autoregressive Layout Generation

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Project page: https://fenghora.github.io/LaviGen-Page/

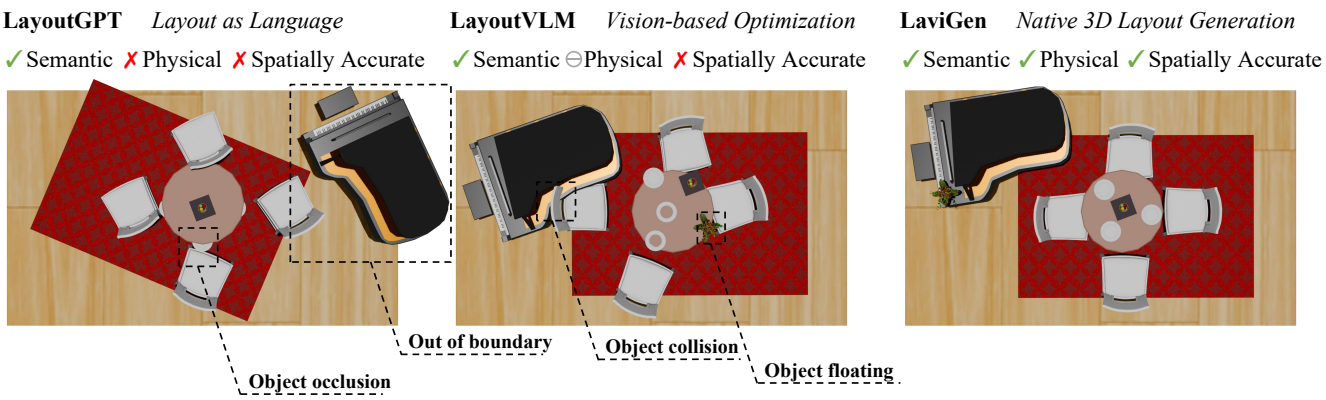


Figure 1. *LaviGen* generates layouts that are both physically plausible and semantically coherent from only 3D objects and instructions, whereas two baselines struggle. Our framework achieves that by leveraging the 3D prior knowledge of generative models to perform generation directly in the native 3D space.

Abstract

We introduce LaviGen, a framework that repurposes 3D generative models for 3D layout generation. Unlike previous methods that infer object layouts from textual descriptions, LaviGen operates directly in the native 3D space, formulating layout generation as an autoregressive process that explicitly models geometric relations and physical constraints among objects, producing coherent and physically plausible 3D scenes. To further enhance this process, we propose an adapted 3D diffusion model that integrates scene, object, and instruction information and employs a dual-guidance self-rollout distillation mechanism to improve efficiency and spatial accuracy. Extensive experiments on the LayoutVLM benchmark show LaviGen achieves superior 3D layout generation performance, with 19% higher physical plausibility than the state of the art and 65% faster computation. Our code is publicly available at https://github.com/fenghora/LaviGen.

1. Introduction

Generating coherent 3D scene layouts [24, 36, 38, 50, 52, 57, 62, 64, 66, 76, 92–94, 99, 105] is essential for creating realistic and interactive VR/AR environments. It aims to arrange objects in semantically consistent and physics-compliant configurations, such as placing chairs around a table instead of against walls. A central challenge, therefore, lies in effectively encoding the geometric distributions describing spatial relationships and semantic dependencies among objects in 3D space.

Early approaches [66] rely on limited 3D scene data with insufficient knowledge about real spatial relationships, and thus lead to physically implausible scene layouts. Recent methods [18, 52, 62, 93, 94, 105] such as Layout-

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GPT [18] treat layout as language in a structured, JSON-like format, which can be generated by large language models (LLMs) [1, 9, 16, 77]. While rich language prior from the LLMs have been employed, the absence of physical modeling often leads to spatially inconsistent layouts, resulting in object collisions, inter-penetrations, or floating. To address this limitation, LayoutVLM [73] leverages visual signals to indirectly supervise layout generation, enhancing the visual plausibility of the resulting scenes. However, image-level supervision is computationally costly and lacks a fundamental understanding of 3D spatial structures.

Inspired by the observation that scene layout is a special type of geometric distribution, we pose the question: *Can layout generation be learned directly from geometric distributions of 3D scenes?* Recent progress in 3D generative modeling [36, 64, 89] has made this feasible, offering powerful 3D priors that encode spatial relationships and geometric distributions. A key challenge, consequently, is how to harness these 3D priors, inherently providing spatial coherence, to enable layout generation, completion, and editing, tasks that are beyond the reach of previous text-based methods. In this context, we repurpose 3D generative models for autoregressive layout generation: leveraging their built-in geometric priors about common spatial layouts, the model places objects sequentially to produce updated scene states that inherently satisfy physically plausible spatial arrangements. Compared to monolithic scene generation, where injecting all object conditions at once can destabilize the generation process, the autoregressive paradigm provides greater controllability and inherently supports object addition and removal.

However, building a 3D generative model into an autoregressive layout generation system is non-trivial. On the one hand, the generative model must simultaneously perceive and learn to align with both the global space of the scene geometry and the object's own canonical space. On the other hand, autoregressive generation inherently suffers from exposure bias [34]; when generating long sequences, this approach inevitably introduces accumulated spatial errors.

To address these challenges, we propose *LaviGen*, a framework that repurposes 3D generative models for autoregressive layout generation. Given an initial scene and an object, *LaviGen* encodes them and integrates semantic information, then generates an updated scene by placing the object in a semantically consistent manner within the native 3D space. In addition, a post-training strategy is proposed to mitigate exposure bias. *LaviGen* introduces a dual-guidance self-rollout distillation strategy that combines scene-level holistic guidance with step-wise scene-object alignment supervision, mitigating error accumulation in long-sequence generation and improving spatial coherence.

As shown in Fig. 2, *LaviGen* leverages the geometric priors of 3D scenes to autoregressively place objects in sequence, achieving more geometrically consistent and plausible layout reasoning while avoiding time-consuming iterative refinement [73]. Extensive experiments on the benchmark proposed by LayoutVLM [73] demonstrate that *LaviGen* outperforms existing layout generation approaches, achieving 19% higher physical plausibility than the state of the art and reducing computational time by roughly 65%. It also supports a broader range of applications, such as layout completion and layout editing, which are difficult to achieve without operating in native 3D space. The main contributions of this work are summarized as follows:

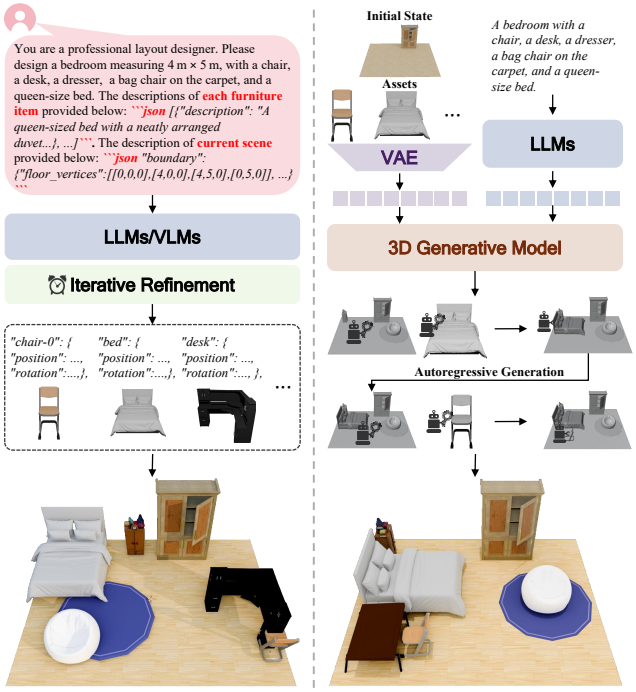
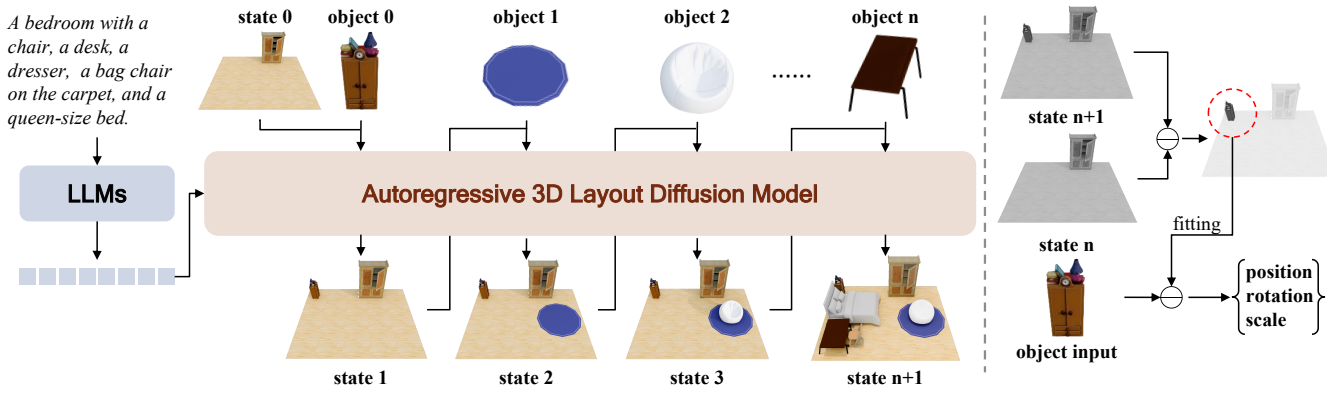


Figure 2. Our layout generation pipeline *versus* existing methods that treat layouts as language or rely on vision-based optimization.

- We propose *LaviGen*, a framework that repurposes a 3D generative model for autoregressive layout generation, enabling layout synthesis directly in the native 3D space.
- Adapted 3D diffusion model and dual-guidance self-rollout distillation to capture environment-object contextual relationships and mitigate exposure bias.
- Experiments show that *LaviGen* achieves superior physical plausibility and generalization in layout synthesis and extends naturally to layout completion and layout editing.

2. Related Work

3D Layout Generation. Generating coherent 3D layouts is a long-standing challenge [24, 50, 52, 62, 64, 66, 92–94, 105]. Early learning-based methods, represented by ATISS [66], employed autoregressive transformers to directly regress object placements. This direct coordinate



(a) Autoregressive Layout Diffusion (b) Object Pose Computation

Figure 3. **Overview of the *LaviGen* framework for autoregressive 3D layout generation.** (a) LaviGen formulates layout generation as an autoregressive process. Specifically, conditioned on LLM-encoded instructions, it takes the current scene state S_i and object O_i to generate the updated state S_{i+1}. (b) The high-fidelity scene is obtained by computing the spatial difference between S_{i+1} and S_i to locate the newly generated region, and fitting the object O_i to derive its spatial parameters.

prediction, however, often neglected geometric semantics, leading to spatial inconsistencies. Subsequent approaches turned to foundation models, initiated by methods that treat layout as a language task [18, 52, 62, 93, 94, 105]. These models leverage LLMs to output structured textual plans, excelling at semantic coherence. A noted limitation, however, is the difficulty in capturing explicit physical constraints, resulting in object collisions or floating artifacts. To address these physical inconsistencies, LayoutVLM [73] introduced 2D visual supervision, using rendered images and differentiable optimization to refine poses. While this improves plausibility, this 2D supervision is not fully holistic for complex 3D interactions and introduces computationally expensive optimization. Both paradigms operate in non-native representations. In contrast, LaviGen formulates the task as a native 3D autoregressive process, operating directly in 3D space to explicitly model geometric relations and physical constraints from the ground up.

3D Generative Models. Recent development in diffusion models [28, 71] and large-scale 3D datasets [12, 13] has greatly advanced the field of 3D generation [5, 6, 14, 20, 25, 26, 30, 35, 39, 44, 45, 47, 53, 55, 56, 58, 59, 61, 63, 65, 69, 74, 78–82, 84, 86, 88, 90, 95, 100, 101, 103]. A series of research [33, 37, 58, 59, 68, 74, 78, 80, 82, 90] generates multi-view images and then reconstructs 3D assets, but two-stage bias often degrades geometric and textural fidelity. A growing body of work [7, 15, 45, 46, 48, 49, 51, 75, 85–87, 89, 100, 102, 103] has explored native 3D diffusion architectures, typically combining a variational autoencoder [42] for latent encoding with a diffusion transformer (DiT) [67] for structured denoising in 3D space. Exhibiting high 3D fidelity and structural consistency, these models learn rich spatial relationships from large-scale 3D data, offering strong geometric priors that underpin our accurate and physically consistent layout generation.

Autoregressive Diffusion and Distillation. The sequential placement of objects naturally frames layout generation as a long-sequence autoregressive generation problem. However, conventional diffusion models with bidirectional attention perform poorly on such autoregressive tasks [22, 29, 43, 66]. Concurrently, the autoregressive generation process inherently suffers from exposure bias [34], leading to accumulated errors [3, 4, 23, 70, 98]. To alleviate this, Diffusion Forcing [21, 31, 40] trains models to denoise tokens conditioned on ground-truth context with independently sampled noise levels. Self Forcing [10, 32, 34, 54, 91] further improves stability by performing autoregressive rollouts during training, conditioning on the model's own outputs. We adopt a similar distillation-based autoregressive mechanism to enhance both efficiency and stability.

3. Methodology

As illustrated in Fig. 3, *LaviGen* is a unified framework that repurposes a pretrained 3D generative model for language-conditioned 3D layout synthesis. Leveraging structured 3D priors, our framework ensures spatial coherence and physical plausibility, substantially reducing object collisions and boundary violations. We begin by revisiting structured 3D latent models [89] in Sec. 3.1, which form the geometric backbone of our approach. We then detail the overall generative pipeline in Sec. 3.2, the autoregressive layout generation mechanism in Sec. 3.3, and the dual-guidance self-rollout scheme at post-training in Sec. 3.4, which jointly enable efficient, accurate, and editable 3D layout synthesis.

3.1. Preliminary

Structured 3D generative models. The 3D prior module in *LaviGen* draws inspiration from structured 3D latent

diffusion models TRELLIS [89], which typically generate 3D assets through a two-stage denoising process that first reconstructs coarse spatial structures and then refines fine-grained geometry and appearance. *LaviGen* retains only the structure-level generation stage, predicting sparse voxel occupancies to model the spatial organization of objects and capture physically and semantically plausible spatial relationships. Each 3D asset is represented by a set of voxel-indexed local latent codes

$$\mathcal{Z} = \{z_p \mid p \in \mathcal{P}\}, \tag{1}$$

where $\mathcal{P}$ denotes the set of active voxel positions near the object surface, and each $z_p \in \mathbb{R}^d$ is the local latent attached to voxel p. This structured representation allows for accurate modeling of 3D space. For generation, TRELLIS adopts Flow Matching models, which add noise ϵ to clean data samples x_0 through $x(t) = (1-t)x_0 + t\epsilon$ over time step t. The reverse dynamics are expressed as a time-dependent vector field $v(x,t) = \nabla_t x$, which is learned via a neural approximation v_θ by minimizing the flow matching loss:

$$\mathcal{L} = \mathbb{E}_{t,x_0,\epsilon} \left\| v_\theta(x,t) - (\epsilon - x_0) \right\|_2^2 . \tag{2}$$

3.2. LaviGen for 3D Layout Generation

In this work, we introduce *LaviGen*, a 3D generative model for autoregressive layout generation that fundamentally differs from prior approaches [18, 66, 73, 93, 105] predicting the spatial coordinates of objects from textual descriptions. Our framework directly models the spatial configuration of objects in the native 3D space, enabling the generation of layouts that are both physically plausible and semantically coherent with textual descriptions.

As illustrated in Fig. 3, given the current state S_i, a target object O_i, and the corresponding layout instruction, *LaviGen* generates a physically and semantically plausible updated layout S_{i+1} directly within the 3D space, which then serves as the initial state for subsequent generation steps. Specifically, the layout instruction is encoded into a conditioning vector c for subsequent generation steps. Subsequently, during each generation step, the current state S_i and the target object O_i are encoded and concatenated with a noise latent, which are then fed into an autoregressive 3D layout diffusion model. The model then performs denoising conditioned on c through cross-attention, producing the updated scene state S_{i+1}. This new state is then combined with the next object O_{i+1}, and the process is repeated in an autoregressive manner to synthesize a coherent 3D layout sequence. Finally, to reconstruct a high-fidelity 3D scene, we extract surface points from the high-resolution voxel occupancy decoded by the VAE. We then downsample the original furniture meshes and register them to the extracted surface points via Iterative Closest Point, estimating optimal rotation, scale, and translation parameters through least-squares fitting. The aligned objects are then placed into the generated layout to obtain the final scene.

3.3. Autoregressive 3D Layout Diffusion

The autoregressive layout generation enables the model to reconstruct the given scene S and integrate a new object O into it, producing spatially coherent and physically plausible layouts. To this end, we design two key components: an **adapted 3D diffusion model** and an **identity-aware embedding module**. These designs collectively enable the model to understand the current scene context and the 3D geometric properties of objects, generating physically plausible and semantically coherent layouts in native 3D space.

Architecture Adaptation. To enable object placement, we adapt the original 3D diffusion model by integrating scene, object, and noisy latents into a unified latent space to capture structured geometric relationships. Specifically, during training, the scene S and object O are first encoded into latent representations s and o, respectively, where $s, o \in \mathbb{R}^{N \times d}$, with $N = H \times W \times L$ representing the total number of latent voxels in a grid of height H, width W, and length L, while d denotes the feature dimension of each voxel. The latent representation of the target scene x_0 is perturbed with a randomly sampled noise $\epsilon \in \mathbb{R}^{N \times d}$, matching the shape of s and o, and then concatenated with them and processed together with the textual embedding c to guide denoising through semantic conditioning.

The overall training objective is

$$\mathcal{L} = \mathbb{E}_{t,x_0,s,o,c,\epsilon} \left\| v_\theta(x,s,o,c,t) - (\epsilon - x_0) \right\|_2^2 . \tag{3}$$

Identity-aware Positional Embedding. Although the adapted 3D diffusion model enables interactions among scene, object, and latent tokens, distinguishing the current scene state from newly added objects remains challenging. To mitigate this, an identity-aware embedding is introduced to explicitly encode the source identity of each token. Concretely, we assign identical positional encodings to the noisy latent x and the state s, reflecting their shared spatial coordinates, while the object o receives a distinct encoding to preserve its individual geometric semantics. This is implemented by extending the standard Rotary Position Embedding (RoPE) [72] with an additional identity flag f indicating the source of each token. After concatenating the input $[x, s, o]$, each token is associated with a voxel at position (f, h, w, l), where $f = 0$ for the noisy latent x and state s, and $f = 1$ for the object o. The spatial coordinates (h, w, l) denote the voxel's position within its respective latent grid. The complex-valued positional frequencies are computed as

$$\Phi(f,h,w,l) = [\,\phi_f(f);\, \phi_h(h);\, \phi_w(w);\, \phi_l(l)\,], \tag{4}$$

where $\phi_f(f)$ encodes the latent source identity and ϕ_h, ϕ_w, ϕ_l follow the standard RoPE for spatial positions.

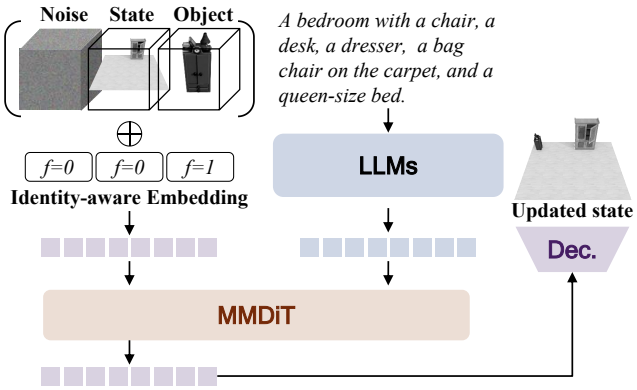


Figure 4. The overview of the **adapted 3D diffusion model**. The encoded scene state and object are concatenated with the noisy latent and, together with the identity-aware embedding, fed into the multimodal diffusion transformer for noise prediction. The denoised output is then decoded to produce the updated scene state.

By embedding identity information in this manner, the model distinguishes different latent streams while preserving spatial alignment, thus enabling precise semantic disentanglement and geometry-consistent reasoning.

3.4. Post-Training via Dual-Guidance Self-Rollout

While the autoregressive paradigm enables progressive scene composition, it suffers from exposure bias: the model is trained on ground-truth context but must condition on its own imperfect outputs at inference, causing errors to accumulate as collisions and implausible placements. We mitigate this via self-rollout post-training, where the student autoregressively rolls out layouts from its own predictions during training, supervised by a holistic teacher for scene-level quality and a step-wise teacher for per-object accuracy.

Self-Rollout Mechanism. Inspired by Self-Forcing [34], we introduce a self-rollout distillation framework. In contrast to Teacher Forcing, which conditions on ground-truth context S_{i-1}, our method performs an autoregressive rollout during training, where the student G_θ conditions on its own generated layout S^θ_{i-1}:

$$\text{Teacher Forcing:} \quad S^\theta_i = G_\theta(S_{i-1}, O_i, c), \tag{5}$$

$$\text{Self-Rollout:} \quad S^\theta_i = G_\theta(S^\theta_{i-1}, O_i, c), \tag{6}$$

where $i = 1, \ldots, n$ and $S^\theta_0 = S_0$. By replacing ground-truth conditioning with self-generated context, we force the model to encounter and learn to recover from its own errors, effectively bridging the train–test distribution gap and reducing error accumulation. We instantiate $\mathcal{L}_{DM}$ as distribution matching distillation [96], minimizing the reverse KL divergence via score distillation with a learned critic model. A key distinction from prior self-rollout methods in video generation [34] is that our autoregressive states are *cumulative*, as each S_i implicitly encodes all previously placed objects, unlike video frames which are independent rendering targets. Consequently, per-frame supervision strategies used in video are insufficient here—errors in early placements propagate into all subsequent states, demanding supervision that addresses both the global scene quality and individual object placements. This motivates our dual-guidance design below. We provide pseudocode in the supplementary material.

Holistic Guidance. Given this cumulative structure, a natural first approach is to supervise only the final scene S^θ_n. We use the bidirectional base model as a global planner $p_{\mathcal{T}_S}$, conditioned on text c, to provide holistic supervision over S^θ_n generated from $\mathcal{C} = (S_0, \{O_i\}_{i=1}^n, c)$:

$$\mathcal{L}_{holistic} = \mathcal{L}_{DM}\left(p_\theta(S_n|\mathcal{C}) \,||\, p_{\mathcal{T}_S}(S_n|c)\right) \tag{7}$$

However, this alone proved insufficient, since supervision at only the terminal state provides no intermediate correction, and the teacher $p_{\mathcal{T}_S}$, not conditioned on object O_i, offers scene-level but not object-placement guidance.

Step-Wise Guidance. To provide dense, object-aware supervision at every step, we use the causal autoregressive model from Sec. 3.3 as a per-step teacher $p_{\mathcal{T}_P}$. Conditioned on O_i, it provides corrective signals at each step i based on the student's imperfect context. Let $\mathcal{C}_i = (S^\theta_{i-1}, O_i, c)$:

$$\mathcal{L}_{step} = \sum_{i=1}^{N} \mathcal{L}_{DM}\left(p_\theta(S_i|\mathcal{C}_i) \,||\, p_{\mathcal{T}_P}(S_i|\mathcal{C}_i)\right) \tag{8}$$

Our final dual-guidance objective combines both terms with equal weights:

$$\mathcal{L}_{dual} = \mathcal{L}_{holistic} + \mathcal{L}_{step}. \tag{9}$$

Concretely, G_θ is updated to align with the teacher score $s_\mathcal{T}$ via:

$$\nabla_\theta \mathcal{L}_{dual} \approx \mathbb{E}_{x_t,t}[(s_\mathcal{T}(x_t,t) - s_\psi(x_t,t))\nabla_\theta x_0], \tag{10}$$

where s_ψ is the critic score approximating the student distribution. By training on self-rolled-out sequences, the student is directly exposed to its own error distribution; the holistic and step-wise teachers then provide complementary corrective signals at the scene and object level, respectively. Full mathematical derivations and pseudocode are provided in the supplementary material.

4. Experiments

4.1. Setting

Implementation Details. We implemented *LaviGen* with an architecture inspired by Trellis [89], reusing its structured variational autoencoder module, and trained the

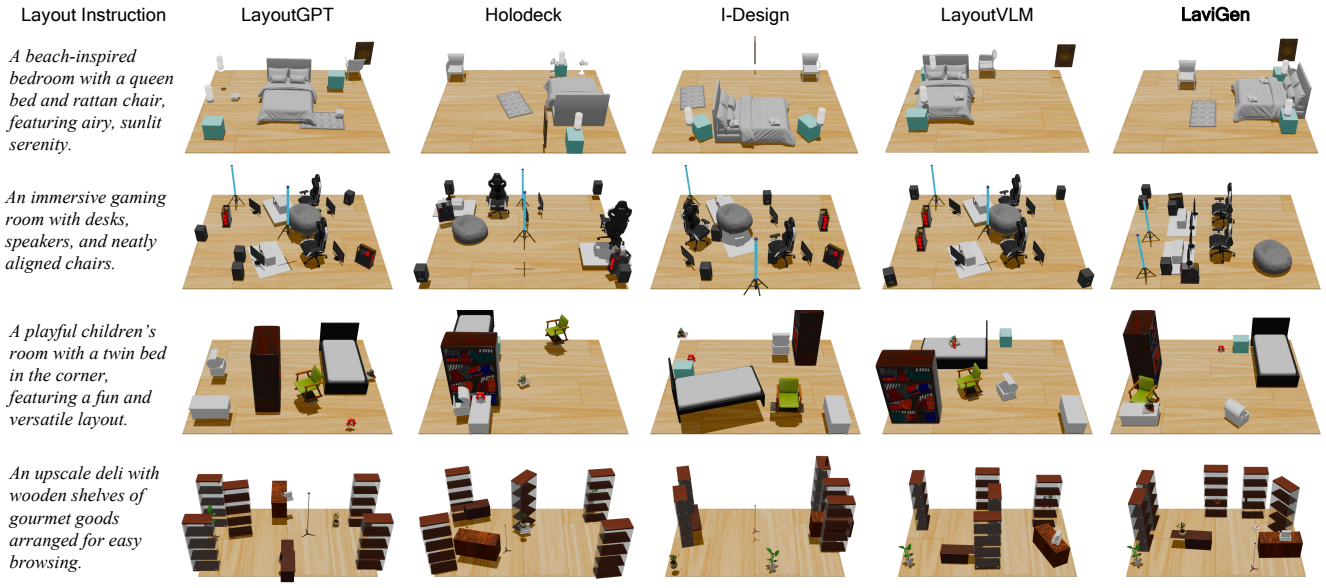


Figure 5. **Qualitative comparison of text-to-3D-layout generation.** *LaviGen* produces physically plausible and spatially coherent layouts aligned with text prompts, effectively avoiding common failure cases such as object collisions (e.g., "gaming room") and floating artifacts (e.g., "children's room", "deli") observed in baselines.

model from scratch using a three-stage training process. First, we replace the original text encoder with Qwen2.5-VL-7B-Instruct [2], keeping its parameters frozen, and train the remaining model as the base bidirectional 3D generative model. Next, we adopted an autoregressive generation paradigm to train the teacher model, which serves as the foundation for subsequent distillation and efficient inference. Finally, we conduct dual-guidance self-rollout distillation: the teacher is distilled into a few-step student, which is then trained with our hybrid objective using the bidirectional model as the holistic teacher and the causal model as the step-wise teacher. For autoregressive ordering, the Qwen-VL encoder derives a semantic object sequence from the input instruction during training, while inference supports either this learned order or user-defined sequences such as bottom-up placement. Our model employs a 3B-parameter DiT and converges stably without delicate hyperparameter tuning, with 20 epochs for base training, 20k steps for teacher fine-tuning, and 5k steps for distillation.

Datasets. For the first training stage, we use the same dataset as Trellis [89], which contains approximately 500K high-quality 3D assets collected from four public datasets: Objaverse-XL [11], ABO [8], 3D-FUTURE [19], and HSSD [41]. For each 3D model, we first render its corresponding images and then use GPT-4o [77] to generate semantically rich annotations. In the second and third stages, we train our model on two large-scale scene datasets, 3D-FRONT [19] and InternScenes [104], comprising about 15K high-quality layout scenes with well-structured spatial arrangements. For fair comparison, we follow LayoutVLM [73] to evaluate *LaviGen* on the same benchmark.

Metrics. We follow LayoutVLM [73] and evaluate 3D layouts in terms of physical plausibility, semantic alignment, and computational efficiency. Physical plausibility is quantified by the *Collision-Free (CF)* and *In-Boundary (IB)* scores, ensuring that objects are non-overlapping and remain within scene boundaries. Semantic alignment is assessed using *Positional (Pos.)* and *Rotational (Rot.) coherency*, which measure the consistency between the generated layout and the textual prompt. For layouts without ground truth, GPT-4o [77] provides semantic ratings from both top-down and side views. We also report the *Physically-Grounded Semantic Alignment (PSA)* score introduced by LayoutVLM, which combines semantic relevance with physical feasibility. Finally, we include the average inference time (T) to evaluate computational efficiency. As LayoutVLM [73] exhibits notably higher latency with increasing object counts, we report results on layouts with 8–10 objects. All metrics except T are normalized to [0,100], where higher values indicate better performance; lower T *(s)* indicates faster inference.

4.2. Main Result

Baselines. We compare our method with several state-of-the-art approaches for text-driven layout generation. LayoutGPT [18] relies on a large language model to directly produce JSON-based layouts. Holodeck [93] and I-Design [105] enhance this process with iterative optimiza-

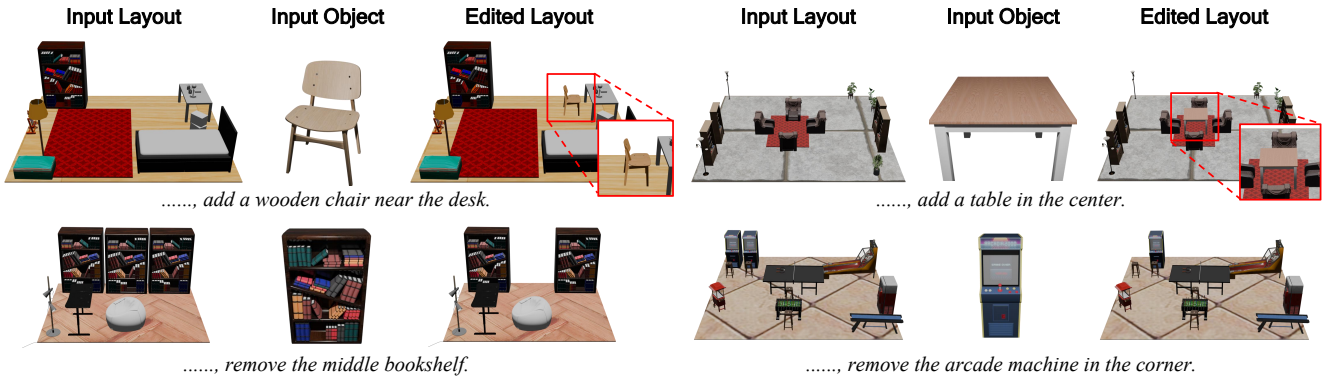


Figure 6. **Qualitative results for layout editing.** *LaviGen's* unified framework supports context-aware modifications, including object insertion (top row) and removal (bottom row). By operating directly in 3D space, the model performs edits that are spatially coherent and semantically consistent with the surrounding context, and enables direct manipulation of 3D layouts, where previous methods struggle.

tion for layout refinement, while LayoutVLM [73] further incorporates visual cues to improve generation quality.

Qualitative and Quantitative Comparison. As shown in Fig. 5, by operating directly in 3D space, *LaviGen* accurately models inter-object relations and produces physically plausible arrangements with minimal collisions or floating artifacts, even in cluttered scenes like a gaming room where other baselines [18, 73, 93, 105] often fail. Examining existing baselines, LayoutGPT [18] generates layouts that are semantically coherent but frequently suffers from collisions, out-of-bound placements, and floating objects. Meanwhile, Holodeck [93] struggles with arranging large objects, which negatively affects its performance. Through iterative optimization, I-Design [105] partially mitigates these issues. LayoutVLM [73] takes a step further by leveraging rendered views to effectively address out-of-bound placements, though collisions and floating objects remain problematic.

Quantitative results in Tab. 1 align with these observations. *LaviGen* achieves the best *CF* and *IB* scores, demonstrating remarkable physical plausibility. LayoutGPT [18] excels in semantic coherence but overlooks physical constraints, resulting in frequent collisions and boundary violations. Holodeck [93] performs poorly on geometric metrics, particularly the *IB* score, as small parameter variations for large objects often cause severe placement errors. I-Design [105] stands as the strongest text-only method, yet its iterative optimization incurs substantial computational cost. LayoutVLM [73], benefiting from visual supervision, performs relatively well overall, though physical plausibility remains suboptimal and rendering introduces additional overhead.

These limitations stem from excessive information compression: representing objects solely with bounding boxes fails to capture fine-grained spatial interactions, and while LayoutVLM partially compensates with visual cues, it is constrained by the trade-off between rendered views and computational cost. These findings underscore the advantage of generating layouts directly in native 3D space, where complete geometric information supports physically plausible and semantically coherent arrangements.

Table 1. **Main quantitative comparison and ablation study.** The top section compares *LaviGen* against state-of-the-art baselines. The bottom section ablates the key components of our model, validating their progressive contributions to the final performance.

Methods	Physical		Semantic		PSA ↑	T (s) ↓
	CF ↑	IB ↑	Pos. ↑	Rot. ↑		
LayoutGPT [18]	83.8	24.2	**80.8**	**78.0**	16.6	**21.3**
Holodeck [93]	77.8	8.1	62.8	55.6	5.6	58.2
I-Design [105]	76.8	34.3	68.3	62.8	18.0	179.2
LayoutVLM [73]	81.8	94.9	77.5	73.2	58.8	75.5
base model	75.6	64.8	45.1	44.7	16.7	145.7
+ id-aware emb.	89.1	96.8	68.8	66.5	71.4	144.1
+ $\mathcal{L}_{holistic}$.	79.5	81.9	61.4	58.7	59.7	24.5
+ $\mathcal{L}_{step}$. **(full)**	**97.3**	**98.6**	76.9	77.1	**78.8**	24.3

User Study. To further evaluate our model, we conducted a user study with 43 participants, each answering 10 questions, yielding 430 responses. For each question, participants selected the best model based on physical plausibility, semantic consistency, and overall quality. Tab. 2 shows that *LaviGen* excels in physical plausibility and overall quality while maintaining comparable semantic consistency.

4.3. Applications

Layout Completion. Layout completion refers to the task of generating a complete layout for a partially specified scene, given layout instructions and 3D assets. This scenario is common, as incomplete or unlabeled 3D layouts often result from limited annotations or missing metadata, making robust completion crucial for downstream tasks. However, language-based approaches [18, 93, 105] strug-

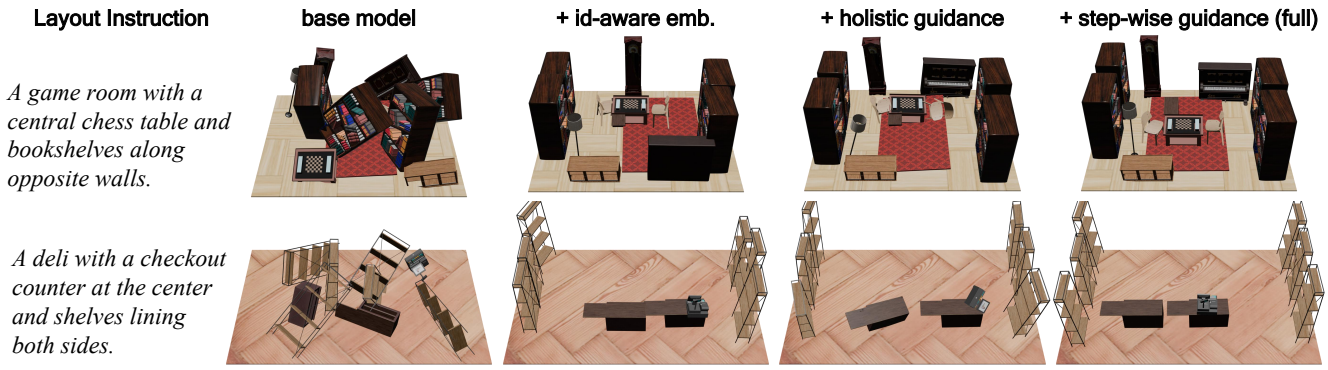


Figure 7. **Qualitative ablation study for *LaviGen*.** We show the progressive improvement from the base model (left) to the full model (right). The baseline produces cluttered layouts with severe collisions, while adding the identity-aware embedding yields a more plausible distribution but still suffers collisions from exposure bias. Distillation with holistic guidance yields inaccurate object fitting and severe inversion errors for small objects. In contrast, the full model generates physically plausible and semantically coherent layouts.

Table 2. User study results for layout generation.

Methods	Phys. Plaus. ↑	Sem. Consist. ↑	Ovr. Qual. ↑
LayoutGPT [18]	16.0	38.8	7.9
LayoutVLM [73]	31.9	27.7	36.5
LaviGen	52.1	33.5	55.6

gle in these cases, as they rely on textual cues rather than 3D spatial understanding. In contrast, *LaviGen* successfully completes this task by operating directly in 3D space, placing objects into the current scene with physical plausibility and semantic coherence. This capability makes it well-suited for applications such as robotic perception, AR/VR environment generation, and autonomous navigation, where comprehensive textual annotations are often unavailable.

Layout Editing. Layout editing is also a practically important capability, allowing users to interact directly with 3D scenes. With a simple adjustment, *LaviGen* can perform high-quality edits, as shown in Fig. 6. To enable this capability, we modify the training paradigm by swapping the autoregressive targets, allowing the model to remove objects from a scene and regenerate them in a context-aware manner. This formulation enables object removal, insertion, and replacement within a single framework, allowing the model to perform edits that are spatially coherent and semantically consistent with the surrounding context. By operating directly in 3D space without relying on textual cues, *LaviGen* further demonstrates strong generalization and practicality for real-world layout modification tasks.

4.4. Ablation Study

To assess the contribution of each component, we conduct ablation studies starting from the base 3D generative model and progressively adding the identity-aware embedding, holistic guidance, and step-wise guidance. The qualitative and quantitative results are presented in Fig. 7 and Tab. 1. Initially, the baseline fails to correctly interpret object–scene relationships, producing cluttered and semantically inconsistent layouts with severe collisions. Adding the identity-aware embedding improves layout coherence, but exposure bias still introduces extraneous points between objects, causing collisions; without distillation, inference also remains slow due to excessive steps. When distilled with holistic guidance, the generation time is greatly reduced, but object-fitting accuracy suffers, particularly for small objects, leading to errors in rotation prediction and noticeable flipping artifacts. Finally, with the introduction of step-wise guidance, i.e., the proposed *LaviGen*, the model produces physically plausible and semantically consistent layouts. These results demonstrate the advantage of generating layouts directly in native 3D space and validate the effectiveness of the proposed framework.

5. Conclusion

We introduced *LaviGen*, a framework for autoregressive 3D layout generation that operates directly in native 3D space. Unlike prior approaches that treat layout as language, LaviGen leverages the geometric priors encoded in 3D generative models, enabling physically plausible and semantically coherent layout generation. We introduce an adapted autoregressive 3D layout diffusion model, fully modeling the spatial relationships between the current scene and input objects to generate an updated scene. To mitigate exposure bias in long-sequence generation, we further propose dual-guidance self-rollout distillation, improving training stability and physical fidelity. Experiments demonstrate that LaviGen achieves superior spatial accuracy and efficiency across baselines, highlighting how its 3D generative paradigm provides a principled foundation for geometry-aware, semantically controllable scene generation.

Repurposing 3D Generative Model for Autoregressive Layout Generation

Supplementary Material

6. Implementation Details

6.1. Base 3D Generative Model

To equip our system with a strong and expressive 3D prior, we begin by training a base 3D generative model. Our design follows the state-of-the-art structured 3D generative framework TRELLIS [89], particularly its structure-level generation stage, which predicts sparse voxel occupancies to capture the spatial organization of objects and to model physically and semantically plausible scene layouts. Concretely, we reuse the structured variational autoencoder of TRELLIS as the backbone, providing a compact and expressive representation of 3D structures. To further enhance the model's ability to interpret complex semantic layouts, we adopt Qwen2.5-VL-7B-Instruct [2] as the text encoder, ensuring rich cross-modal grounding. The design of our flow transformer builds upon the architecture of Qwen-Image [83] and integrates the Multimodal Diffusion Transformer (MMDiT) [17], thereby enabling unified modeling of text and 3D representations within a single Transformer framework. Within each block of the MMDiT, we incorporate a novel positional encoding mechanism, Multimodal Scalable RoPE (MSRoPE), designed to provide consistent and scale-robust positional representations across both modalities. This formulation enables effective multimodal fusion while preserving stable positional semantics for text and scalable spatial modeling for 3D latent representations.

The detailed experimental settings are as follows. We represent the 3D scene using a 64^3 voxel grid, which is used across all training stages as well as during inference. For training, we adopt classifier-free guidance (CFG) [27] with a drop rate of 0.1 and use the AdamW optimizer [60] with a learning rate of 1×10^{-4}. The model is trained for 400K steps on 16 A100 GPUs (80GB) with a batch size of 16 per GPU. At inference time, the CFG strength is set to 3 and 50 sampling steps are used.

6.2. Teacher Model

With a strong 3D prior established, the next stage focuses on applying it to layout generation. To preserve the spatial knowledge already acquired by the base model, we minimize modifications to the original architecture. Specifically, as illustrated in Sec. 3.3, the teacher model builds upon the base 3D generative model by jointly taking the current scene state and the target object as input. To enable the model to distinguish between the scene and objects while facilitating faster convergence, we further introduce an identity-aware positional embedding. Together, these designs allow the teacher model to achieve comprehensive modeling of the geometric relationships between the scene and objects. The experimental setup is largely consistent with that of the first stage, with the main differences being a reduced training length of 100K steps and a learning rate of 5×10^{-5}.

6.3. Post-Training via Dual-Guidance Self-Rollout

To mitigate the exposure bias inherent in autoregressive generation, we employ a dual-guidance self-rollout strategy, as summarized in Algorithm 1. This stage distills a pre-trained, few-step student generator following the methodology detailed in the main paper Sec. 3.4. Below, we specify the network components and hyperparameters used in this process.

The distillation framework comprises four distinct models. The Student Model G_θ is an efficient, few-step 3D layout diffusion model, initialized via distillation from the Autoregressive Teacher Model trained in Sec. 6.2. The Holistic Teacher $p_{\mathcal{T}_S}$ provides the final supervision signal $\mathcal{L}_{holistic}$ and is implemented using the frozen Base 3D Generative Model described in Sec. 6.1. The Step-Wise Teacher $p_{\mathcal{T}_P}$ provides intermediate corrective signals $\mathcal{L}_{step}$ utilizing the frozen Autoregressive Teacher Model detailed in Sec. 6.2. Finally, to implement the Distribution Matching Distillation (DMD) loss, we employ a trainable Critic Model f_ψ. This critic is initialized with the same architecture and weights as the Base 3D Generative Model, i.e., the Holistic Teacher, and is trained to approximate the score function of the student's generated data distribution.

Training Hyperparameters. We perform post-training with a batch size of 1 given the sequential, memory-intensive nature of the self-rollout process. The Student Model G_θ is optimized using AdamW with a learning rate of 2×10^{-6}, $(\beta_1, \beta_2) = (0.0, 0.999)$, and weight decay of 0.01. The Critic Network f_ψ is optimized separately using AdamW with a learning rate of 5×10^{-7}, $(\beta_1, \beta_2) = (0.0, 0.999)$, and weight decay of 0.01. To stabilize score estimation, we use a Generator/Critic update ratio of 1:5 (i.e., five critic updates per student update). For teacher score computation $s_{\mathcal{T}}$, Classifier-Free Guidance (CFG) is applied with a scale of 3.0.

Loss Function Formulation. Our dual-guidance objective $\mathcal{L}_{dual} = \mathcal{L}_{holistic} + \mathcal{L}_{step}$ is formulated using Distribution Matching Distillation [97]. This objective minimizes the reverse Kullback-Leibler divergence by leveraging the

score difference between the student (approximated by the critic f_ψ) and the teacher. The gradient for the student G_θ is derived as follows:

$$\nabla_\theta \mathcal{L}_{dual} \approx \mathbb{E}_{x_t,t}\left[(s_\mathcal{T}(x_t,t) - s_\psi(x_t,t))\nabla_\theta x_0\right] \quad (11)$$

where $x_0 = G_\theta(x_t, t, \mathcal{C}_i)$ denotes the clean layout predicted by the student. Here, $s_\mathcal{T}$ represents the score function of the fixed teacher (either $p_{\mathcal{T}_S}$ for $\mathcal{L}_{holistic}$ or $p_{\mathcal{T}_P}$ for $\mathcal{L}_{step}$), and s_ψ is the score estimated by the critic. The critic is concurrently trained to approximate the student's score using a standard denoising objective.

7. Additional Results

7.1. Scalability to More Objects

We evaluate on 8–10 objects for fair comparison against existing baselines. Thanks to self-rollout distillation, LaviGen inherently supports a "train short, test long" paradigm and can handle scenes with more than 20 objects, as shown in Fig. 8.

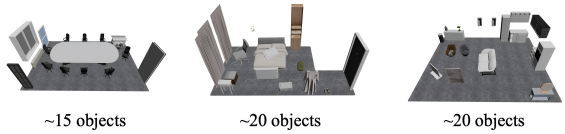


Figure 8. Qualitative results for long-sequence generation with more than 20 objects.

7.2. Generalizability Across Backbones

Our framework is not tied to a specific 3D generative backbone. To validate this, we apply LaviGen to TRELLIS [89] using its original CLIP text encoder, without our additionally trained Qwen encoder. As shown in Fig. 9, the model maintains high physical plausibility and semantic coherence, confirming that our recipe successfully transfers across different base architectures without relying on large-scale training infrastructure.

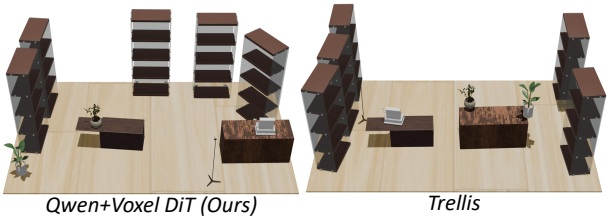


Figure 9. Generalization across different 3D generation backbones.

7.3. Generation Diversity

LaviGen naturally supports diverse outputs via stochastic sampling. Given the same input instruction, the model generates varied yet plausible layouts, as illustrated in Fig. 10.

Algorithm 1 Dual-Guidance Self-Rollout Distillation

Require: Denoise timesteps $\{t_1, \ldots, t_T\}$, number of objects N
Require: Student generator G_θ, step-wise teacher $p_{\mathcal{T}_P}$, holistic teacher $p_{\mathcal{T}_S}$
Require: Initial state S_0, object sequence $\{O_i\}_{i=1}^N$, text prompt c

loop
  $S_{\text{ctx}} \leftarrow S_0$, $S_{\text{outputs}} \leftarrow []$
  Sample $s \sim \text{Uniform}(1, \ldots, T)$
  for $i = 1, \ldots, N$ **do**
    $\mathcal{C}_i \leftarrow (S_{\text{ctx}}, O_i, c)$
    Initialize $z_{t_T} \sim \mathcal{N}(0, I)$
    for $j = T, \ldots, s$ **do**
      if $j == s$ **then**
        Enable gradient computation
        $\hat{S}_0 \leftarrow G_\theta(z_{t_j}; t_j, \mathcal{C}_i)$
        S_{outputs}.append($\hat{S}_0$)
        $S_{\text{ctx}} \leftarrow \hat{S}_0$.detach()
      else
        Disable gradient computation
        $\hat{S}_0 \leftarrow G_\theta(z_{t_j}; t_j, \mathcal{C}_i)$
        Sample $\epsilon \sim \mathcal{N}(0, I)$
        $z_{t_{j-1}} \leftarrow \Psi(\hat{S}_0, \epsilon, t_{j-1})$
      end if
    end for
  end for

  $\mathcal{L}_{step} \leftarrow 0$, $S_{\text{ctx}} \leftarrow S_0$
  for $i = 1, \ldots, N$ **do**
    $\hat{S}_i \leftarrow S_{\text{outputs}}[i]$
    $\mathcal{C}_i \leftarrow (S_{\text{ctx}}, O_i, c)$
    $\mathcal{L}_{step} \leftarrow \mathcal{L}_{step} + \mathcal{L}_{\text{DMD}}(\hat{S}_i; p_{\mathcal{T}_P}, \mathcal{C}_i)$
    $S_{\text{ctx}} \leftarrow \hat{S}_i$.detach()
  end for
  $\hat{S}_N \leftarrow S_{\text{outputs}}$.last()
  $\mathcal{L}_{holistic} \leftarrow \mathcal{L}_{\text{DMD}}(\hat{S}_N; p_{\mathcal{T}_S}, c)$
  $\mathcal{L}_{dual} \leftarrow \mathcal{L}_{step} + \mathcal{L}_{holistic}$
  Update θ via $\nabla_\theta \mathcal{L}_{dual}$
end loop

8. Limitations and Future Work.

Although *LaviGen* gains strong geometric distribution modeling capability from operating in the native 3D space, several limitations remain. First, due to constraints in model capacity and computational resources, we adopt a 64^3 3D grid resolution. While generally adequate for most objects, this resolution becomes insufficient for small instances, leading to mismatches in subsequent spatial coordinate computations. To address this issue, our future work will explore more efficient computation strategies for higher-resolution voxel grids and investigate denser 3D representations capable of supporting higher spatial resolutions and capturing finer spatial details. Additionally, as shown in Tab. 1, the semantic consistency of the generated layouts remains suboptimal. We attribute this primarily to the scarcity of high-quality annotations, particularly for layouts



Figure 10. Diverse layout results generated from the same input instruction.

with complex spatial configurations or object arrangements. In future work, we will enhance our annotation pipeline to collect and process additional high-quality labeled data, and explore more advanced text-conditioning mechanisms to further improve the robustness and semantic reliability of *LaviGen*.

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References

[1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023. 2

[2] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report. In *arXiv*, 2025. 6, 1

[3] Boyuan Chen, Diego Martí Monsó, Yilun Du, Max Simchowitz, Russ Tedrake, and Vincent Sitzmann. Diffusion forcing: Next-token prediction meets full-sequence diffusion. *Advances in Neural Information Processing Systems*, 37:24081–24125, 2024. 3

[4] Guibin Chen, Dixuan Lin, Jiangping Yang, Chunze Lin, Juncheng Zhu, Mingyuan Fan, Hao Zhang, Sheng Chen, Zheng Chen, Chengchen Ma, et al. Skyreels-v2: Infinite-length film generative model. *arXiv preprint arXiv:2504.13074*, 2025. 3

[5] Sijin Chen, Xin Chen, Anqi Pang, Xianfang Zeng, Wei Cheng, Yijun Fu, Fukun Yin, Zhibin Wang, Jingyi Yu, Gang Yu, et al. Meshxl: Neural coordinate field for generative 3d foundation models. *Advances in Neural Information Processing Systems*, 37:97141–97166, 2024. 3

[6] Yiwen Chen, Tong He, Di Huang, Weicai Ye, Sijin Chen, Jiaxiang Tang, Xin Chen, Zhongang Cai, Lei Yang, Gang Yu, Guosheng Lin, and Chi Zhang. Meshanything: Artist-created mesh generation with autoregressive transformers, 2024. 3

[7] Yiwen Chen, Zhihao Li, Yikai Wang, Hu Zhang, Qin Li, Chi Zhang, and Guosheng Lin. Ultra3d: Efficient and high-fidelity 3d generation with part attention, 2025. 3

[8] Jasmine Collins, Shubham Goel, Kenan Deng, Achleshwar Luthra, Leon Xu, Erhan Gundogdu, Xi Zhang, Tomas F Yago Vicente, Thomas Dideriksen, Himanshu Arora, et al. Abo: Dataset and benchmarks for real-world 3d object understanding. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 21126–21136, 2022. 6

[9] Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025. 2

[10] Justin Cui, Jie Wu, Ming Li, Tao Yang, Xiaojie Li, Rui Wang, Andrew Bai, Yuanhao Ban, and Cho-Jui Hsieh. Self-forcing++: Towards minute-scale high-quality video generation. *arXiv preprint arXiv:2510.02283*, 2025. 3

[11] Matt Deitke, Ruoshi Liu, Matthew Wallingford, Huong Ngo, Oscar Michel, Aditya Kusupati, Alan Fan, Christian Laforte, Vikram Voleti, Samir Yitzhak Gadre, et al. Objaverse-xl: A universe of 10m+ 3d objects. *Advances in Neural Information Processing Systems*, 36:35799–35813, 2023. 6

[12] Matt Deitke, Dustin Schwenk, Jordi Salvador, Luca Weihs, Oscar Michel, Eli VanderBilt, Ludwig Schmidt, Kiana Ehsani, Aniruddha Kembhavi, and Ali Farhadi. Objaverse: A universe of annotated 3d objects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13142–13153, 2023. 3

[13] Matt Deitke, Ruoshi Liu, Matthew Wallingford, Huong Ngo, Oscar Michel, Aditya Kusupati, Alan Fan, Christian Laforte, Vikram Voleti, Samir Yitzhak Gadre, et al. Objaverse-xl: A universe of 10m+ 3d objects. *Advances in Neural Information Processing Systems*, 36, 2024. 3

[14] Junting Dong, Qi Fang, Zehuan Huang, Xudong Xu, Jingbo Wang, Sida Peng, and Bo Dai. Tela: Text to layer-wise 3d clothed human generation. In *European Conference on Computer Vision*, pages 19–36. Springer, 2025. 3

[15] Shaocong Dong, Lihe Ding, Xiao Chen, Yaokun Li, Yuxin Wang, Yucheng Wang, Qi Wang, Jaehyeok Kim, Chenjian Gao, Zhanpeng Huang, et al. From one to more: Contextual part latents for 3d generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 8230–8240, 2025. 3

[16] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The

llama 3 herd of models. *arXiv e-prints*, pages arXiv–2407, 2024. 2

[17] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first international conference on machine learning*, 2024. 1

[18] Weixi Feng, Wanrong Zhu, Tsu-jui Fu, Varun Jampani, Arjun Akula, Xuehai He, Sugato Basu, Xin Eric Wang, and William Yang Wang. Layoutgpt: Compositional visual planning and generation with large language models. *Advances in Neural Information Processing Systems*, 36: 18225–18250, 2023. 1, 2, 3, 4, 6, 7, 8

[19] Huan Fu, Rongfei Jia, Lin Gao, Mingming Gong, Binqiang Zhao, Steve Maybank, and Dacheng Tao. 3d-future: 3d furniture shape with texture. *International Journal of Computer Vision*, 129(12):3313–3337, 2021. 6

[20] Daoyi Gao, Yawar Siddiqui, Lei Li, and Angela Dai. Meshart: Generating articulated meshes with structure-guided transformers. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 618–627, 2025. 3

[21] Kaifeng Gao, Jiaxin Shi, Hanwang Zhang, Chunping Wang, Jun Xiao, and Long Chen. Ca2-vdm: Efficient autoregressive video diffusion model with causal generation and cache sharing. In *ICML*. PMLR, 2025. 3

[22] Songwei Ge, Thomas Hayes, Harry Yang, Xi Yin, Guan Pang, David Jacobs, Jia-Bin Huang, and Devi Parikh. Long video generation with time-agnostic vqgan and time-sensitive transformer. In *ECCV*, 2022. 3

[23] Yuchao Gu, Weijia Mao, and Mike Zheng Shou. Long-context autoregressive video modeling with next-frame prediction. *arXiv preprint arXiv:2503.19325*, 2025. 3

[24] Zeqi Gu, Yin Cui, Zhaoshuo Li, Fangyin Wei, Yunhao Ge, Jinwei Gu, Ming-Yu Liu, Abe Davis, and Yifan Ding. Artiscene: Language-driven artistic 3d scene generation through image intermediary. *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2891–2901, 2025. 1, 2

[25] Zekun Hao, David W. Romero, Tsung-Yi Lin, and Ming-Yu Liu. Meshtron: High-fidelity, artist-like 3d mesh generation at scale, 2024. 3

[26] Zexin He, Tengfei Wang, Xin Huang, Xingang Pan, and Ziwei Liu. Neural lightrig: Unlocking accurate object normal and material estimation with multi-light diffusion. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 26514–26524, 2025. 3

[27] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *arXiv*, 2022. 1

[28] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020. 3

[29] Wenyi Hong, Ming Ding, Wendi Zheng, Xinghan Liu, and Jie Tang. Cogvideo: Large-scale pretraining for text-to-video generation via transformers. In *ICLR*, 2023. 3

[30] Yicong Hong, Kai Zhang, Jiuxiang Gu, Sai Bi, Yang Zhou, Difan Liu, Feng Liu, Kalyan Sunkavalli, Trung Bui, and Hao Tan. Lrm: Large reconstruction model for single image to 3d. *arXiv preprint arXiv:2311.04400*, 2023. 3

[31] Jinyi Hu, Shengding Hu, Yuxuan Song, Yufei Huang, Mingxuan Wang, Hao Zhou, Zhiyuan Liu, Wei-Ying Ma, and Maosong Sun. Acdit: Interpolating autoregressive conditional modeling and diffusion transformer. *arXiv preprint arXiv:2412.07720*, 2024. 3

[32] Junchao Huang, Xinting Hu, Boyao Han, Shaoshuai Shi, Zhuotao Tian, Tianyu He, and Li Jiang. Memory forcing: Spatio-temporal memory for consistent scene generation on minecraft. *arXiv preprint arXiv:2510.03198*, 2025. 3

[33] Xiufeng Huang, Ka Chun Cheung, Runmin Cong, Simon See, and Renjie Wan. Stereo-gs: Multi-view stereo vision model for generalizable 3d gaussian splatting reconstruction. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pages 9822–9831, 2025. 3

[34] Xun Huang, Zhengqi Li, Guande He, Mingyuan Zhou, and Eli Shechtman. Self forcing: Bridging the train-test gap in autoregressive video diffusion. *arXiv preprint arXiv:2506.08009*, 2025. 2, 3, 5

[35] Zehuan Huang, Hao Wen, Junting Dong, Yaohui Wang, Yangguang Li, Xinyuan Chen, Yan-Pei Cao, Ding Liang, Yu Qiao, Bo Dai, et al. Epidiff: Enhancing multi-view synthesis via localized epipolar-constrained diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9784–9794, 2024. 3

[36] Zehuan Huang, Yuan-Chen Guo, Xingqiao An, Yunhan Yang, Yangguang Li, Zi-Xin Zou, Ding Liang, Xihui Liu, Yan-Pei Cao, and Lu Sheng. Midi: Multi-instance diffusion for single image to 3d scene generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025. 1, 2

[37] Zehuan Huang, Yuan-Chen Guo, Haoran Wang, Ran Yi, Lizhuang Ma, Yan-Pei Cao, and Lu Sheng. Mv-adapter: Multi-view consistent image generation made easy. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16377–16387, 2025. 3

[38] Zhening Huang, Xiaoyang Wu, Fangcheng Zhong, Hengshuang Zhao, Matthias Nießner, and Joan Lasenby. Litereality: Graphics-ready 3d scene reconstruction from rgb-d scans, 2025. 1

[39] Moritz Ibing, Gregor Kobsik, and Leif Kobbelt. Octree transformer: Autoregressive 3d shape generation on hierarchically structured sequences. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2698–2707, 2023. 3

[40] Yang Jin, Zhicheng Sun, Ningyuan Li, Kun Xu, Hao Jiang, Nan Zhuang, Quzhe Huang, Yang Song, Yadong Mu, and Zhouchen Lin. Pyramidal flow matching for efficient video generative modeling. In *ICLR*, 2025. 3

[41] Mukul Khanna, Yongsen Mao, Hanxiao Jiang, Sanjay Haresh, Brennan Shacklett, Dhruv Batra, Alexander Clegg, Eric Undersander, Angel X Chang, and Manolis Savva. Habitat synthetic scenes dataset (hssd-200): An analysis of 3d scene scale and realism tradeoffs for objectgoal navigation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16384–16393, 2024. 6

[42] Diederik P Kingma. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013. 3

[43] Dan Kondratyuk, Lijun Yu, Xiuye Gu, Jose Lezama, Jonathan Huang, Grant Schindler, Rachel Hornung, Vighnesh Birodkar, Jimmy Yan, Ming-Chang Chiu, et al. Videopoet: A large language model for zero-shot video generation. In *ICML*, 2024. 3

[44] Songlin Li, Despoina Paschalidou, and Leonidas Guibas. Pasta: Controllable part-aware shape generation with autoregressive transformers. *arXiv preprint arXiv:2407.13677*, 2024. 3

[45] Weiyu Li, Jiarui Liu, Rui Chen, Yixun Liang, Xuelin Chen, Ping Tan, and Xiaoxiao Long. Craftsman: High-fidelity mesh generation with 3d native generation and interactive geometry refiner. *arXiv preprint arXiv:2405.14979*, 2024. 3

[46] Weiyu Li, Jiarui Liu, Hongyu Yan, Rui Chen, Yixun Liang, Xuelin Chen, Ping Tan, and Xiaoxiao Long. Craftsman3d: High-fidelity mesh generation with 3d native diffusion and interactive geometry refiner. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 5307–5317, 2025. 3

[47] Weiyu Li, Xuanyang Zhang, Zheng Sun, Di Qi, Hao Li, Wei Cheng, Weiwei Cai, Shihao Wu, Jiarui Liu, Zihao Wang, Xiao Chen, Feipeng Tian, Jianxiong Pan, Zeming Li, Gang Yu, Xiangyu Zhang, Daxin Jiang, and Ping Tan. Step1x-3d: Towards high-fidelity and controllable generation of textured 3d assets, 2025. 3

[48] Yangguang Li, Zi-Xin Zou, Zexiang Liu, Dehu Wang, Yuan Liang, Zhipeng Yu, Xingchao Liu, Yuan-Chen Guo, Ding Liang, Wanli Ouyang, et al. Triposg: High-fidelity 3d shape synthesis using large-scale rectified flow models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025. 3

[49] Yangguang Li, Zi-Xin Zou, Zexiang Liu, Dehu Wang, Yuan Liang, Zhipeng Yu, Xingchao Liu, Yuan-Chen Guo, Ding Liang, Wanli Ouyang, et al. Triposg: High-fidelity 3d shape synthesis using large-scale rectified flow models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025. 3

[50] Chenguo Lin and Yadong Mu. Instructscene: Instruction-driven 3d indoor scene synthesis with semantic graph prior. In *International Conference on Learning Representations (ICLR)*, 2024. 1, 2

[51] Yuchen Lin, Chenguo Lin, Panwang Pan, Honglei Yan, Yiqiang Feng, Yadong Mu, and Katerina Fragkiadaki. Partcrafter: Structured 3d mesh generation via compositional latent diffusion transformers, 2025. 3

[52] Lu Ling, Chen-Hsuan Lin, Tsung-Yi Lin, Yifan Ding, Yu Zeng, Yichen Sheng, Yunhao Ge, Ming-Yu Liu, Aniket Bera, and Zhaoshuo Li. Scenethesis: Combining language and visual priors for 3d scene generation. *arXiv preprint arXiv:2505.02836*, 2025. 1, 2, 3

[53] Anran Liu, Cheng Lin, Yuan Liu, Xiaoxiao Long, Zhiyang Dou, Hao-Xiang Guo, Ping Luo, and Wenping Wang. Part123: part-aware 3d reconstruction from a single-view image. In *ACM SIGGRAPH 2024 Conference Papers*, pages 1–12, 2024. 3

[54] Kunhao Liu, Wenbo Hu, Jiale Xu, Ying Shan, and Shijian Lu. Rolling forcing: Autoregressive long video diffusion in real time, 2025. 3

[55] Minghua Liu, Ruoxi Shi, Linghao Chen, Zhuoyang Zhang, Chao Xu, Xinyue Wei, Hansheng Chen, Chong Zeng, Jiayuan Gu, and Hao Su. One-2-3-45++: Fast single image to 3d objects with consistent multi-view generation and 3d diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10072–10083, 2024. 3

[56] Minghua Liu, Chao Xu, Haian Jin, Linghao Chen, Mukund Varma T, Zexiang Xu, and Hao Su. One-2-3-45: Any single image to 3d mesh in 45 seconds without per-shape optimization. *Advances in Neural Information Processing Systems*, 36, 2024. 3

[57] Xinhang Liu, Yu-Wing Tai, and Chi-Keung Tang. Agentic 3d scene generation with spatially contextualized vlms. *arXiv preprint arXiv:2505.20129*, 2025. 1

[58] Yuan Liu, Cheng Lin, Zijiao Zeng, Xiaoxiao Long, Lingjie Liu, Taku Komura, and Wenping Wang. Syncdreamer: Generating multiview-consistent images from a single-view image. *arXiv preprint arXiv:2309.03453*, 2023. 3

[59] Xiaoxiao Long, Yuan-Chen Guo, Cheng Lin, Yuan Liu, Zhiyang Dou, Lingjie Liu, Yuexin Ma, Song-Hai Zhang, Marc Habermann, Christian Theobalt, et al. Wonder3d: Single image to 3d using cross-domain diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9970–9980, 2024. 3

[60] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *arXiv*, 2019. 1

[61] Shuqi Lu, Haowei Lin, Lin Yao, Zhifeng Gao, Xiaohong Ji, Weinan E, Linfeng Zhang, and Guolin Ke. Uni-3dar: Unified 3d generation and understanding via autoregression on compressed spatial tokens. *Arxiv*, 2025. 3

[62] Yongsen Mao, Junhao Zhong, Chuan Fang, Jia Zheng, Rui Tang, Hao Zhu, Ping Tan, and Zihan Zhou. Spatiallm: Training large language models for structured indoor modeling. In *Advances in Neural Information Processing Systems*, 2025. 1, 2, 3

[63] Quan Meng, Lei Li, Matthias Nießner, and Angela Dai. Lt3sd: Latent trees for 3d scene diffusion. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 650–660, 2025. 3

[64] Yanxu Meng, Haoning Wu, Ya Zhang, and Weidi Xie. Scenegen: Single-image 3d scene generation in one feedforward pass. In *arXiv*, 2025. 1, 2

[65] Charlie Nash, Yaroslav Ganin, SM Ali Eslami, and Peter Battaglia. Polygen: An autoregressive generative model of 3d meshes. In *International conference on machine learning*, pages 7220–7229. PMLR, 2020. 3

[66] Despoina Paschalidou, Amlan Kar, Maria Shugrina, Karsten Kreis, Andreas Geiger, and Sanja Fidler. Atiss: Autoregressive transformers for indoor scene synthesis. *Advances in neural information processing systems*, 34: 12013–12026, 2021. 1, 2, 3, 4

[67] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF In-*

ternational Conference on Computer Vision, pages 4195–4205, 2023. 3

[68] Yansong Qu, Shaohui Dai, Xinyang Li, Yuze Wang, You Shen, Shengchuan Zhang, and Liujuan Cao. Deocc-1-to-3: 3d de-occlusion from a single image via self-supervised multi-view diffusion. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 8677–8685, 2026. 3

[69] Barbara Roessle, Norman Müller, Lorenzo Porzi, Samuel Rota Bulò, Peter Kontschieder, Angela Dai, and Matthias Nießner. L3dg: Latent 3d gaussian diffusion. In *SIGGRAPH Asia 2024 Conference Papers*, pages 1–11, 2024. 3

[70] Sand-AI. Magi-1: Autoregressive video generation at scale, 2025. 3

[71] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020. 3

[72] Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568: 127063, 2024. 4

[73] Fan-Yun Sun, Weiyu Liu, Siyi Gu, Dylan Lim, Goutam Bhat, Federico Tombari, Manling Li, Nick Haber, and Jiajun Wu. Layoutvlm: Differentiable optimization of 3d layout via vision-language models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 29469–29478, 2025. 2, 3, 4, 6, 7, 8

[74] Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. Lgm: Large multi-view gaussian model for high-resolution 3d content creation. In *European Conference on Computer Vision*, pages 1–18. Springer, 2025. 3

[75] Jiaxiang Tang, Ruijie Lu, Zhaoshuo Li, Zekun Hao, Xuan Li, Fangyin Wei, Shuran Song, Gang Zeng, Ming-Yu Liu, and Tsung-Yi Lin. Efficient part-level 3d object generation via dual volume packing, 2025. 3

[76] Xiang Tang, Ruotong Li, and Xiaopeng Fan. Towards geometric and textural consistency 3d scene generation via single image-guided model generation and layout optimization. *arXiv preprint arXiv:2507.14841*, 2025. 1

[77] OpenAI teams. Gpt-4o system card. In *arXiv*, 2024. 2, 6

[78] Vikram Voleti, Chun-Han Yao, Mark Boss, Adam Letts, David Pankratz, Dmitry Tochilkin, Christian Laforte, Robin Rombach, and Varun Jampani. Sv3d: Novel multi-view synthesis and 3d generation from a single image using latent video diffusion. In *European Conference on Computer Vision*, pages 439–457. Springer, 2025. 3

[79] Zhengyi Wang, Jonathan Lorraine, Yikai Wang, Hang Su, Jun Zhu, Sanja Fidler, and Xiaohui Zeng. Llama-mesh: Unifying 3d mesh generation with language models, 2024.

[80] Zhengyi Wang, Yikai Wang, Yifei Chen, Chendong Xiang, Shuo Chen, Dajiang Yu, Chongxuan Li, Hang Su, and Jun Zhu. Crm: Single image to 3d textured mesh with convolutional reconstruction model. In *European conference on computer vision*, pages 57–74. Springer, 2024. 3

[81] Si-Tong Wei, Rui-Huan Wang, Chuan-Zhi Zhou, Baoquan Chen, and Peng-Shuai Wang. Octgpt: Octree-based multiscale autoregressive models for 3d shape generation. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*, pages 1–11, 2025.

[82] Hao Wen, Zehuan Huang, Yaohui Wang, Xinyuan Chen, and Lu Sheng. Ouroboros3d: Image-to-3d generation via 3d-aware recursive diffusion. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 21631–21641, 2025. 3

[83] Chenfei Wu, Jiahao Li, Jingren Zhou, Junyang Lin, Kaiyuan Gao, Kun Yan, Sheng ming Yin, Shuai Bai, Xiao Xu, Yilei Chen, Yuxiang Chen, Zecheng Tang, Zekai Zhang, Zhengyi Wang, An Yang, Bowen Yu, Chen Cheng, Dayiheng Liu, Deqing Li, Hang Zhang, Hao Meng, Hu Wei, Jingyuan Ni, Kai Chen, Kuan Cao, Liang Peng, Lin Qu, Minggang Wu, Peng Wang, Shuting Yu, Tingkun Wen, Wensen Feng, Xiaoxiao Xu, Yi Wang, Yichang Zhang, Yongqiang Zhu, Yujia Wu, Yuxuan Cai, and Zenan Liu. Qwen-image technical report. In *arXiv*, 2025. 1

[84] Kailu Wu, Fangfu Liu, Zhihan Cai, Runjie Yan, Hanyang Wang, Yating Hu, Yueqi Duan, and Kaisheng Ma. Unique3d: High-quality and efficient 3d mesh generation from a single image. *Advances in Neural Information Processing Systems*, 37:125116–125141, 2024. 3

[85] Ruiqi Wu, Xinjie Wang, Liu Liu, Chunle Guo, Jiaxiong Qiu, Chongyi Li, Lichao Huang, Zhizhong Su, and Ming-Ming Cheng. Dipo: Dual-state images controlled articulated object generation powered by diverse data, 2025. 3

[86] Shuang Wu, Youtian Lin, Feihu Zhang, Yifei Zeng, Jingxi Xu, Philip Torr, Xun Cao, and Yao Yao. Direct3d: Scalable image-to-3d generation via 3d latent diffusion transformer. *Advances in Neural Information Processing Systems*, 37: 121859–121881, 2024. 3

[87] Shuang Wu, Youtian Lin, Feihu Zhang, Yifei Zeng, Yikang Yang, Yajie Bao, Jiachen Qian, Siyu Zhu, Xun Cao, Philip Torr, and Yao Yao. Direct3d-s2: Gigascale 3d generation made easy with spatial sparse attention, 2025. 3

[88] Zhennan Wu, Yang Li, Han Yan, Taizhang Shang, Weixuan Sun, Senbo Wang, Ruikai Cui, Weizhe Liu, Hiroyuki Sato, Hongdong Li, et al. Blockfusion: Expandable 3d scene generation using latent tri-plane extrapolation. *ACM Transactions on Graphics (TOG)*, 43(4):1–17, 2024. 3

[89] Jianfeng Xiang, Zelong Lv, Sicheng Xu, Yu Deng, Ruicheng Wang, Bowen Zhang, Dong Chen, Xin Tong, and Jiaolong Yang. Structured 3d latents for scalable and versatile 3d generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2025. 2, 3, 4, 5, 6, 1

[90] Jiale Xu, Weihao Cheng, Yiming Gao, Xintao Wang, Shenghua Gao, and Ying Shan. Instantmesh: Efficient 3d mesh generation from a single image with sparse-view large reconstruction models. *arXiv preprint arXiv:2404.07191*, 2024. 3

[91] Shuai Yang, Wei Huang, Ruihang Chu, Yicheng Xiao, Yuyang Zhao, Xianbang Wang, Muyang Li, Enze Xie, Yingcong Chen, Yao Lu, and Song Hanand Yukang Chen. Longlive: Real-time interactive long video generation. In *arxiv*, 2025. 3

[92] Yandan Yang, Baoxiong Jia, Peiyuan Zhi, and Siyuan Huang. Physcene: Physically interactable 3d scene synthesis for embodied ai. In *Proceedings of Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 1, 2

[93] Yue Yang, Fan-Yun Sun, Luca Weihs, Eli VanderBilt, Alvaro Herrasti, Winson Han, Jiajun Wu, Nick Haber, Ranjay Krishna, Lingjie Liu, et al. Holodeck: Language guided generation of 3d embodied ai environments. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16227–16237, 2024. 1, 3, 4, 6, 7

[94] Yixuan Yang, Zhen Luo, Tongsheng Ding, Junru Lu, Mingqi Gao, Jinyu Yang, Victor Sanchez, and Feng Zheng. Llm-driven indoor scene layout generation via scaled human-aligned data synthesis and multi-stage preference optimization. *ArXiv*, abs/2506.07570, 2025. 1, 2, 3

[95] Junliang Ye, Zhengyi Wang, Ruowen Zhao, Shenghao Xie, and Jun Zhu. Shapellm-omni: A native multimodal llm for 3d generation and understanding, 2025. 3

[96] Tianwei Yin, Michaël Gharbi, Taesung Park, Richard Zhang, Eli Shechtman, Fredo Durand, and Bill Freeman. Improved distribution matching distillation for fast image synthesis. *NeurIPS*, 2024. 5

[97] Tianwei Yin, Michaël Gharbi, Richard Zhang, Eli Shechtman, Fredo Durand, William T Freeman, and Taesung Park. One-step diffusion with distribution matching distillation. In *CVPR*, 2024. 1

[98] Tianwei Yin, Qiang Zhang, Richard Zhang, William T Freeman, Fredo Durand, Eli Shechtman, and Xun Huang. From slow bidirectional to fast autoregressive video diffusion models. In *CVPR*, 2025. 3

[99] Huangyue Yu, Baoxiong Jia, Yixin Chen, Yandan Yang, Puhao Li, Rongpeng Su, Jiaxin Li, Qing Li, Wei Liang, Zhu Song-Chun, Tengyu Liu, and Siyuan Huang. Metascenes: Towards automated replica creation for real-world 3d scans. In *Conference on Computer Vision and Pattern Recognition(CVPR)*, 2025. 1

[100] Longwen Zhang, Ziyu Wang, Qixuan Zhang, Qiwei Qiu, Anqi Pang, Haoran Jiang, Wei Yang, Lan Xu, and Jingyi Yu. Clay: A controllable large-scale generative model for creating high-quality 3d assets. *ACM Transactions on Graphics (TOG)*, 43(4):1–20, 2024. 3

[101] Ruowen Zhao, Junliang Ye, Zhengyi Wang, Guangce Liu, Yiwen Chen, Yikai Wang, and Jun Zhu. Deepmesh: Autoregressive artist-mesh creation with reinforcement learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10612–10623, 2025. 3

[102] Wang Zhao, Yan-Pei Cao, Jiale Xu, Yuejiang Dong, and Ying Shan. Assembler: Scalable 3d part assembly via anchor point diffusion. In *Proceedings of the SIGGRAPH Asia 2025 Conference Papers*, pages 1–11, 2025. 3

[103] Zibo Zhao, Wen Liu, Xin Chen, Xianfang Zeng, Rui Wang, Pei Cheng, Bin Fu, Tao Chen, Gang Yu, and Shenghua Gao. Michelangelo: Conditional 3d shape generation based on shape-image-text aligned latent representation. *Advances in Neural Information Processing Systems*, 36, 2024. 3

[104] Weipeng Zhong, Peizhou Cao, Yichen Jin, Li Luo, Wenzhe Cai, Jingli Lin, Hanqing Wang, Zhaoyang Lyu, Tai Wang, Bo Dai, Xudong Xu, and Jiangmiao Pang. Internscenes: A large-scale simulatable indoor scene dataset with realistic layouts. In *arXiv*, 2025. 6

[105] Ata Çelen, Guo Han, Konrad Schindler, Luc Van Gool, Iro Armeni, Anton Obukhov, and Xi Wang. I-design: Personalized llm interior designer. In *Computer Vision – ECCV 2024 Workshops*, 2025. 1, 2, 3, 4, 6, 7